

Math 113 – Abstract Algebra

Homework 4 — my solutions

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All assignments must be **turned in on Gradescope** by the deadline. Late assignments are not accepted.

Have questions? Work with your peers, attend office hours, check out the [Student Learning Center](#), and ask in discussion section!

Problems for credit. Turn in these problems; some subset of them will be graded.

1. Fraleigh Exercise 8.4
2. Fraleigh Exercise 8.8
3. Fraleigh Exercise 8.36
4. Fraleigh Exercise 8.52
5. Fraleigh Exercise 9.10
6. Fraleigh Exercise 9.24
7. Fraleigh Exercise 9.33

Practice problems. The following problems will not be graded, and should not be turned in, but are recommended as good practice problems. Solutions to most odd-numbered exercises are available in the back of the textbook.

- Fraleigh Exercises 8.1, 8.5, 8.17, 8.18, 8.21, 8.35, 8.41, 8.43, 8.53
- Fraleigh Exercises 9.11, 9.13, 9.19, 9.27, 9.29, 9.31, 9.39

Solutions

Problem 1 (Fraleigh Exercise 8.4)

Solution. First write σ and τ as disjoint cycles:

$$\sigma = (1, 3, 4, 5, 6, 2), \quad \tau = (1, 2, 4, 3)(5, 6).$$

Squaring the 6-cycle moves two steps at a time, so $\sigma^2 = (1, 4, 6)(2, 3, 5)$.

Inverting reverses each cycle, so $\sigma^{-2} = (1, 6, 4)(2, 5, 3)$.

Now $\sigma^{-2}\tau$ means apply τ first, then σ^{-2} . Tracing each element:

$$1 \rightarrow 2 \rightarrow 5, \quad 5 \rightarrow 6 \rightarrow 4, \quad 4 \rightarrow 3 \rightarrow 2, \quad 2 \rightarrow 4 \rightarrow 1,$$

so we get $(1, 5, 4, 2)$.

And $3 \rightarrow 1 \rightarrow 6, 6 \rightarrow 5 \rightarrow 3$ gives $(3, 6)$.

Thus

$$\sigma^{-2}\tau = (1, 5, 4, 2)(3, 6).$$

□

Problem 2 (Fraleigh Exercise 8.8)

Solution. σ is a 6-cycle, so it returns to the identity after exactly 6 applications. That is, $|\sigma| = 6$. Hence the exponent reduces modulo 6: $100 = 96 + 4$, so

$$\sigma^{100} = \sigma^4.$$

From Problem 1, $\sigma^2 = (1, 4, 6)(2, 3, 5)$, so squaring once more gives $\sigma^4 = (1, 6, 4)(2, 5, 3)$. □

Problem 3 (Fraleigh Exercise 8.36)

Solution. S_3 works. It's nonabelian since

$$(1, 2)(1, 3) = (1, 3, 2) \quad \text{but} \quad (1, 3)(1, 2) = (1, 2, 3).$$

Now take any proper subgroup H of S_3 . The identity has order 1, the three transpositions have order 2, and the two 3-cycles have order 3. So H can't have order 5 (that would need an element of order 5), and it can't have order 4 either: its three nonidentity elements would all have to be transpositions, but any two distinct transpositions in S_3 multiply to a 3-cycle, which would have to be in H too. So H has order 1, 2, or 3.

If $|H| = 2$, then $H = \{e, g\}$ with $g^2 = e$.

If $|H| = 3$, then $H = \{e, g, g^2\}$.

In either case every element is a power of a single g , so any two elements are powers g^i, g^j , which commute. Thus H is abelian. $\square$

Problem 4 (Fraleigh Exercise 8.52)

Solution. Let $H = \{\rho_a \mid a \in G\}$.

Each ρ_a is a permutation of G . If $\rho_a(x) = \rho_a(y)$, then $xa = ya$, and multiplying both sides by a^{-1} on the right gives $x = y$, so it's one-to-one.

It's onto too, since any y is hit by $y = \rho_a(ya^{-1})$.

Composing two of them:

$$(\rho_a \circ \rho_b)(x) = \rho_a(xb) = (xb)a = x(ba) = \rho_{ba}(x),$$

so $\rho_a \circ \rho_b = \rho_{ba}$.

So the set is closed under composition.

We have that ρ_e is the identity, and $\rho_{a^{-1}}$ is the inverse of ρ_a since $\rho_a \circ \rho_{a^{-1}} = \rho_e$. Thus $H \leq S_G$.

Since composing ρ_a and ρ_b gives ρ_{ba} , we fix the order of the subscripts by using $a \rightarrow \rho_{a^{-1}}$ instead of $a \rightarrow \rho_a$. Define $\phi : G \rightarrow H$ by $\phi(a) = \rho_{a^{-1}}$. Then

$$\phi(ab) = \rho_{(ab)^{-1}} = \rho_{b^{-1}a^{-1}} = \rho_{a^{-1}} \circ \rho_{b^{-1}} = \phi(a)\phi(b).$$

Finally, ϕ is onto by definition, and it's one-to-one since $\phi(a) = \phi(b)$ means $\rho_{a^{-1}} = \rho_{b^{-1}}$, and evaluating at e gives $a^{-1} = b^{-1}$, so $a = b$. So ϕ is an isomorphism, and $G \cong H$. $\square$

Problem 5 (Fraleigh Exercise 9.10)

Solution. We have

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 2 & 6 & 3 & 7 & 4 & 5 & 1 \end{pmatrix}.$$

$1 \rightarrow 8 \rightarrow 1$ gives $(1, 8)$.

2 is fixed.

$3 \rightarrow 6 \rightarrow 4 \rightarrow 3$ gives $(3, 6, 4)$.

And $5 \rightarrow 7 \rightarrow 5$ gives $(5, 7)$.

So

$$\sigma = (1, 8)(3, 6, 4)(5, 7).$$

Finally, $(3, 6, 4) = (3, 4)(3, 6)$, so

$$\sigma = (1, 8)(3, 4)(3, 6)(5, 7).$$

□

Problem 6 (Fraleigh Exercise 9.24)

Solution. The identity is even, being the empty product. The 3-cycles are even as well, since $(1, 2, 3) = (1, 3)(1, 2)$ and $(1, 3, 2) = (1, 2)(1, 3)$. The μ 's are transpositions, hence odd.

Thus

$$A_3 = \{\rho_0, \rho_1, \rho_2\},$$

with multiplication table

	ρ_0	ρ_1	ρ_2
ρ_0	ρ_0	ρ_1	ρ_2
ρ_1	ρ_1	ρ_2	ρ_0
ρ_2	ρ_2	ρ_0	ρ_1

□

Problem 7 (Fraleigh Exercise 9.33)

Solution. Take any odd τ . We need $\tau = \sigma\mu$ for some $\mu \in A_n$, so solve for μ : $\mu = \sigma^{-1}\tau$. Since σ^{-1} is also odd, $\sigma^{-1}\tau$ is even, so $\mu \in A_n$.

Then

$$\sigma\mu = \sigma(\sigma^{-1}\tau) = \tau.$$

so we are done.

□